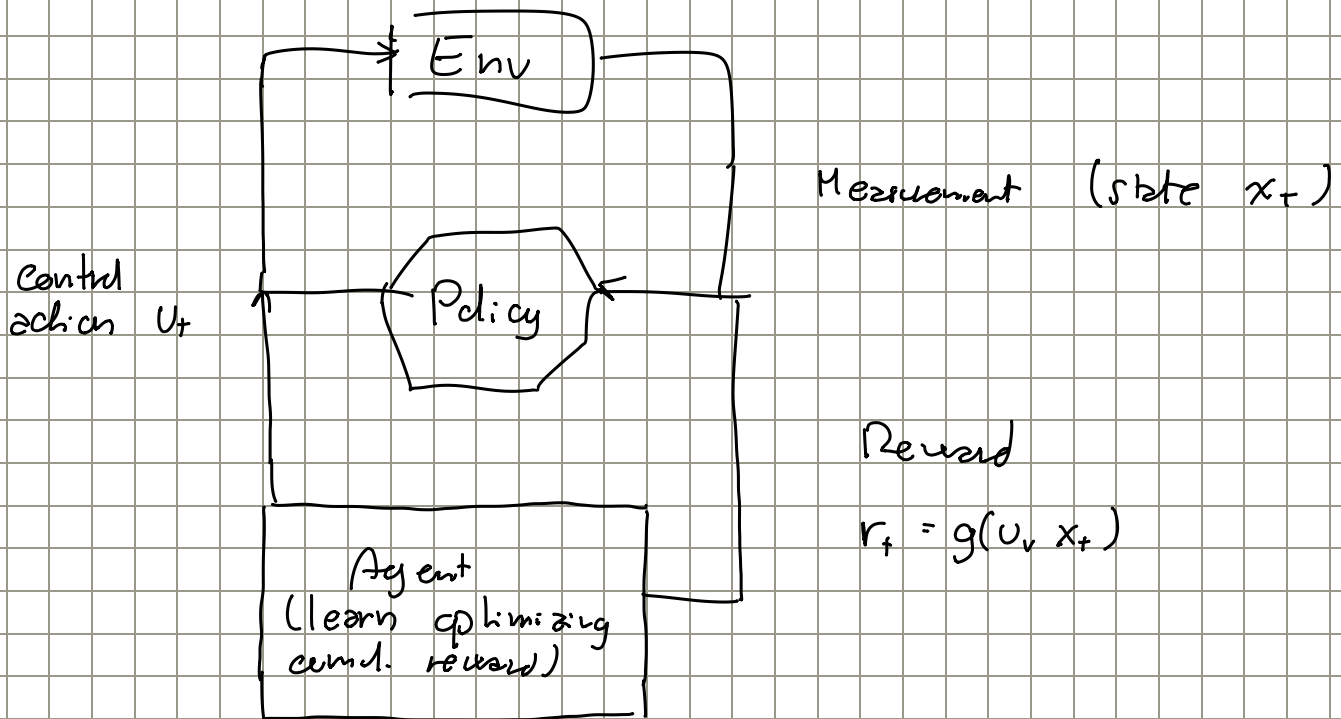


Reinforcement learning

How each of us interact with the world



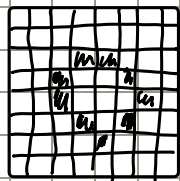
PROBABILITY INTRO

GENERAL START

$x \in X \equiv$ input space

$y \in Y \equiv$ label set $\begin{cases} \text{discrete} \\ \text{continuous} \end{cases}$

$S = \{(x_i, y_i), i = 1, \dots, n\} \equiv \frac{\text{training set}}{(\text{SAMPLE})}$



ex. digital recognition

$y \in \{0, 1, 2, \dots, 9\} \in Y$

$\downarrow$ vectorize
 $x \in \mathbb{R}^{64} \in X$

Data generation assumption

(1) (Simple): $x \sim D_x$, x are random

$\exists f(x)$ s.t. $y_i = f(x_i)$

(2) Realistic

$x \sim D_x$

$y|x \sim D_{y|x}$

$\Rightarrow (x, y) \sim D_{y|x} \cdot D_x$

$$P(y|x) = \frac{P(x, y)}{P(x)}$$

Does it remind you something?

Learning Algorithm

$A(S) \mapsto h \in \underbrace{H}_{\substack{\text{hypothesis} \\ \text{class}}}$

Error function (loss)

$z = (x, y) \longrightarrow l(h, z) \geq 0$

example.

$$\mathcal{Y} = \{0, 1\}$$

$$l(h, z) = \begin{cases} 0 & h(x) = y \\ 1 & h(x) \neq y \end{cases}$$

0-1 loss

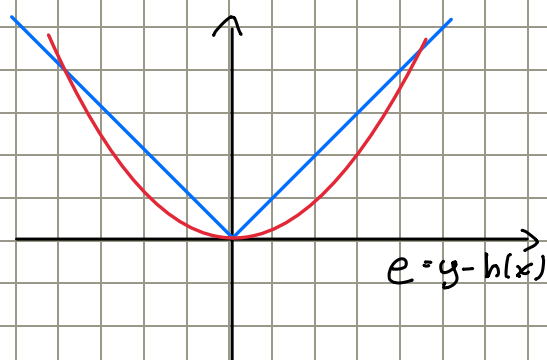
Example: $y \in \mathbb{R}$

$$l(h, z) = (y - h(x))^2$$

QUADRATIC LOSS

$$l(h, z) = |y - h(x)|$$

L_1 loss



Difference?

Ultimate goal

Make

True Risk

Small

$$\mathbb{E}_{z \sim D} [l(h, z)] =: L_D(h)$$

$z \sim D$

TRUE RISK

~~Attempt: to define the alg $A(S)$~~

~~$$A(S) = \arg \min_{h \in H} L_D(h)$$~~

D is unknown to the learner

Back to probability:

$$w \sim D$$

$$m_w = \mathbb{E}_{w \sim D} [w]$$

$$S = \{w_1, w_2, \dots, w_m\}$$

$\hat{m}_w = \text{estimate mean average} = \frac{1}{n} \sum_{i=1}^n w_i \longrightarrow m_w$ Cosa dovrebbe fare: ricordare?
 L.L.N = Law of Large Numbers

He's speaking referring to:

$$E_{z \sim D} [\ell(h, z)] =: L_D(h)$$

GIVEN: $S := \{z_1, \dots, z_n\}$ $z_i := \{x_i, y_i\}$

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \ell[h, z_i] \quad \text{we can evaluate it}$$

↓
 Empirical risk use it as proxy for $L_D(h)$
 with some caveats...

Empirical risk minimization

$$A(S) \in \arg \min_{h \in H} L_S(h)$$

Remark

$$A(S) = \hat{h}(S) \in \arg \min_{h \in H} L_S(h)$$

Ultimate goal

is $L_D(A(S))$ SMALL?

Intuition: L.L.N.

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \ell(h, z_i)$$

We would like to guarantee that:

$$\mathbb{P}[\underbrace{L_D(A(S)) \text{ SMALL}}_{\text{APPROXIMATE}}] \text{ is LARGE}$$

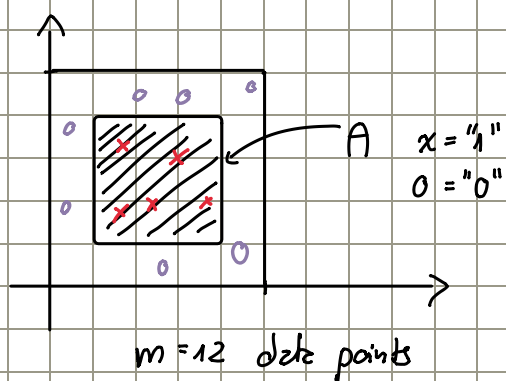
HIGH PROBABLE

Small risk with high prob.

PAC Learning (Probably Approximate Correct)

Warning: this may fail...

Ex.



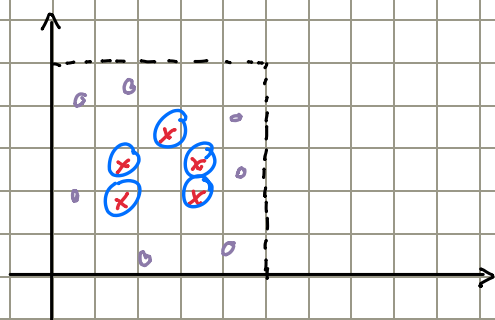
$$x \in [0,1] \times [0,1] \subset \mathbb{R}^2$$

$$x \sim \underbrace{U([0,1] \times [0,1])}_{D_x}$$

$$y = f(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$\uparrow$
 $\{0,1\}$

Learner sees this

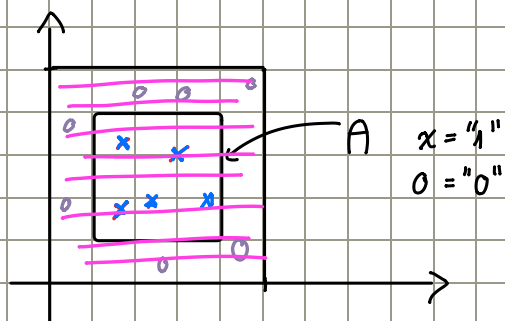


$$\hat{h}_S = \arg \min_{h \in H} L_S(h)$$

$\ell(h, z) = 0-1$ loss (for binary)

$$H = \{ \text{all possible functions } h(x): \underbrace{x}_{\in \mathbb{R}^2} \rightarrow \{0,1\} \}$$

$$\hat{h}(x) = \begin{cases} 1 & \text{if } \exists i \in [m] \text{ s.t. } x = x_i, y_i = 1 \\ 0 & \text{otherwise} \end{cases}$$



function attaches 1 to each point, 0 everywhere else

$$L_0(\hat{h}_S) = E_{z \sim D} [\underbrace{\ell(\hat{h}_S(x), y)}_{\ell(\hat{h}_S, z)}] =$$

$$= \underbrace{1 \cdot \mathbb{P}[\ell(\hat{h}(x), y) = 1]}_{1/2} + 0 \cdot \cancel{\mathbb{P}[\ell(\hat{h}(x), y) = 0]}$$

$= 1/2 \Rightarrow$ this is a clear example of overfitting

Homework (try (*))

$$H = \{ h_B(x) = \begin{cases} 1 & x \in B \\ 0 & x \notin B \end{cases}$$

where B is any
square in $[0,1] \times [0,1]$

